

---

# Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

---

Palaash Gang  
Independent Researcher  
palaashgang@gmail.com

## Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its  
2 refutation by overlapping authors; this paper joins them and locates the edge  
3 of the set that refutation relies on. Conformal PID states long-run coverage for  
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the  
5 deployed threshold inside an admissible radius. The claim was acted on; the  
6 readout’s placement and the corollary’s derivation are credited by name. At the null  
7 scorecaster and the minimal admissible saturator that radius is *tight*. A legal dead  
8 band on the *completed* threshold exists that set at its radius; past it miscoverage is  
9 1.000000 against  $\alpha = 0.1$ : coverage fails outright, not merely its rate. The edge  
10 belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition  
11 is sufficient only. Under the equally legal  $\hat{q} \equiv +b/2$  the failing band covers, as  
12 measured; at  $\hat{q} \equiv -b/2$  both it and partial adjustment at  $w = 0.999$  miscover.

## 13 1 Introduction

14 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,  
15 by the same first and last authors, proves in the same notation that a bounded perturbation of the  
16 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the  
17 joining is what this paper adds.

18 An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round. Forecast  
19 stability names, scores and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle  
20 et al., 2023, Pritularga and Kourntzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying  
21 temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals  
22 on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage  
23 and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is  
24 claimed here.

25 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-  
26 gelopoulos et al., 2023] adds a *scorecaster*  $\hat{q}_{t+1}$  to a saturating error integrator, asking only that it be  
27 predictable and lie in  $[-b/2, b/2]$ : “any function of the past” (Theorem 1). Such a readout is already  
28 quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical  
29 coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, ver-  
30 batim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2]  
31 add any predictable  $d_t$ ,  $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$  with  $\mu_t \in [0, 1]$  predictable, to the deployed threshold;  
32 since  $\hat{q}_t + d_t$  still lies in  $[-b/2, b/2]$  it is itself a legal scorecaster, so the result “has exactly the CPID  
33 error-integration form” and  $|E_T| = o(T)$ . The same generic-slot argument carries AcMCP’s long-run  
34 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv  
35 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:  
36 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”  
37 and lacking “principled guidance” while asserting valid coverage: model selection, not validity. **We**  
38 **claim neither the placement nor the derivation.**

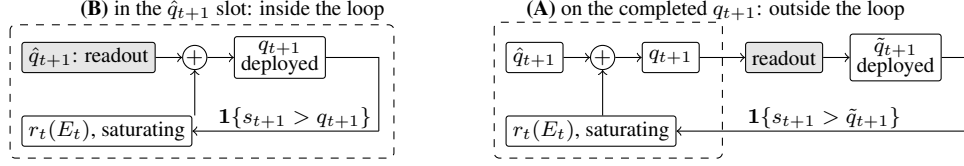


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop  $r_t$  closes. In (B) the loop closes on the integrator’s own output, and both readouts keep  $\{\hat{q}_t\}$  in the convex hull of  $\hat{q}_1$  and the raw scorecaster’s range, inside Corollary 2’s admissible set. In (A) it closes on  $\tilde{q}_{t+1}$ , which it never computed, and nothing keeps the deployed sequence inside that set.

39 **Where Corollary 2 stops, its radius condition is tight.** It is a *retention* result, silent outside that  
40 radius, and the question it leaves is whether the radius is sharp. Section 3 exhibits a legal perturbation  
41 leaving it: the  $L_1$  dead-band family characterised here on the *completed* threshold, whose edge  
42 coincides with the corollary’s radius at the null scorecaster. Past it coverage fails outright, not merely  
43 its rate: the published sufficient condition is necessary there. Appendix C sets both claims beside  
44 four vocabularies for the same movement.

## 45 2 Setup

46 **The producer.** A forecaster fit before  $t$  induces a score  $s_t$ , and a threshold  $q_t$  is emitted before  
47  $y_t$  is revealed. The deployed set is  $C_t = \{y : s_t(x_t, y) \leq q_t\}$  and  $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$ . The  
48 target is long-run coverage,  $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$ , with no model on the data. Conformal PID  
49 [Angelopoulos et al., 2023] manipulates the threshold on the score scale:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left( \sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

50 Angelopoulos et al.’s iteration (5). Its  $r_t$  saturates: condition (4) requires  $|r_t(x)| \geq b$  with the sign of  
51  $x$  once  $|x| \geq c h(t)$ , for  $h$  nonnegative, nondecreasing and sublinear. With  $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$ ,  
52 Proposition 2 gives  $|E_T| \leq c h(T) + 1$ , read on realised scores.

53 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard  
54 readouts, both prior work. A quadratic cost gives linear partial adjustment  $\tilde{q}_{t+1} = (1 - w)\hat{q}_{t+1}^{\text{raw}} + w\tilde{q}_t$   
55 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied  
56 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,  
57 moving only when the target is more than  $\tau$  away, then pulling back exactly  $\tau$  [Constantinides,  
58 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; (A),  
59 Placement A, is measured here. *The dead band ( $L_1$ ) is not a stylistic alternative to the partial-*  
60 *adjustment ( $L_2$ ) form;* it is the family whose failure Section 3 characterises. Neither is safe by  
61 construction:  $L_2$  forfeits Proposition 2’s rate while covering,  $L_1$  past the radius loses coverage, and  
62 at the legal  $\hat{q} \equiv -b/2$  both lose it.

63 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*,  $\mathcal{T}_H =$   
64  $\sum_{t=2}^H |q_t - q_{t-1}|$ , after actuator travel in process control. The arithmetic is not new, and the name is  
65 a convenience rather than a contribution, used here for one reading only: Appendix C names ten for  
66 the same movement across four vocabularies.

67 **Harness.** The harness fixes  $\alpha = 0.1$  and  $b = 2$ . Its saturator  $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$ ,  
68 with  $c = 1$  and  $h(t) = \log(t + 2)$ , meets (4) with equality. The scorecaster is null and legal, reducing  
69 (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated  
70 variable. Neither choice is neutral: (4) is a lower bound met exactly here, and Theorem 1 permits  
71 scorecasters  $b/2$  either side of null, predictable but not required to be constant in  $t$ . The adversary is  
72 specified, not tie-broken. It plays  $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ ,  $\varepsilon = 10^{-9}$ , against the *deployed*  $\tilde{q}_t$ : the  
73  $\varepsilon \downarrow 0$  reading of “sets the score at the threshold”, firing the strict indicator, legal on both sides of  
74 Theorem 1’s  $[-b/2, b/2]$ , so  $b/2 + b/2 = b$  is exactly tight. *That specification names one member*  
75 *of a class:* any adapted  $s_t \in [-b/2, b/2]$  is legal, the constant  $s_t \equiv -b/2$  included. Every number  
76 in Table 1 (App. A) is measured against the one adversary above; Section 3’s two boundaries are  
77 quantified over the class, and they differ.

### 78 3 The edge of the admissible set, located and proved

79 **The retention result is published, and tested from outside.** The radius restated in Section 1 is  
 80 the one Li et al. [2025, Corollary 2] prove, and it *retains*  $|E_T| \leq ch(T) + 1$ : the radius is theirs. A  
 81 downstream partial adjustment of weight  $w$  leaves it; Table 1 (App. A) prices that departure, and five  
 82 of its eleven arms exceed the bound at  $T = 10^6$ . The forfeit is a property of the smoothing *strength*,  
 83 not Placement A, and it buys something: travel falls from 28,311 to 202. A readout on the completed  
 84  $q_{t+1}$  touches neither  $r_t$  nor the integrator’s argument, so it cannot put (4) at risk. It delays a correction  
 85 already computed, so smoothing the output does not damp  $E_t$  but feeds it.

86 **The  $L_1$  dead band leaves the radius, and the edge is a theorem.** The dead band of Section 2 keeps  
 87  $|\hat{q}_{t+1} - q_{t+1}| \leq \tau$  at every round, so “inside Corollary 2” reads  $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$ . Under saturation  
 88 the deployed value is pinned  $\tau$  below the integrator’s ceiling. Outside the radius two levels decide the  
 89 answer and condition (4) names only one of them. Write  $A_t^+ = \sup_x r_t(x)$  and  $A_t^- = \inf_x r_t(x)$  for  
 90 the integrator’s extremes, and  $\Lambda_t^+ = \inf_{x \geq ch(t)} r_t(x)$ ,  $\Lambda_t^- = \sup_{x \leq -ch(t)} r_t(x)$  for the levels it is  
 91 *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly  $\Lambda_t^+ \geq b$  and  
 92  $\Lambda_t^- \leq -b$ ; that  $\Lambda_t^\pm = A_t^\pm$ , that the integrator reaches its extremes where (4) puts them, is a further  
 93 hypothesis and not one (4) supplies. Set

$$\tau^{*\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

94 **Failure, over the whole admissible class.** If  $\tau > \tau^{*+}$  then  $\hat{q}_{t+1} + A_t^+ - \tau < b/2$  at every round,  
 95 so  $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$  makes  $\{\tilde{q} < b/2\}$  absorbing: a release upward lands below  $b/2$ , and a  
 96 hold or a release downward cannot lift the deployed value at all. Against  $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$  the  
 97 indicator is then  $\mathbf{1}\{\tilde{q}_t < b/2\}$  and every round misses. If  $\tau \geq \tau^{*-}$  the constant legal play  $s_t \equiv -b/2$   
 98 makes  $\{\tilde{q} \geq -b/2\}$  absorbing by the mirror inequality, so no round can miss and the realised rate  
 99 is 0 rather than  $\alpha$ . Either way long-run coverage is lost, with  $|E_T|$  linear in  $T$ . Nothing here asks  $\hat{q}$   
 100 to be constant, asks the extremes to be attained anywhere, or asks  $A_t^\pm$  to be finite beyond what the  
 101 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If  $\tau \leq \sigma^+$   
 102 then  $E_{t-1} \geq ch(t-1)$  forces  $q_t \geq b/2 + \tau$  and so  $\tilde{q}_t \geq b/2$ , whence  $\text{err}_t = 0$  against *every* legal  
 103 adversary; if  $\tau < \sigma^-$  the mirror forces  $\text{err}_t = 1$  below  $-ch(t-1)$ . The accumulator cannot then  
 104 leave  $[-ch(t), ch(t)]$  by more than one round, and  $|E_T| \leq ch(T) + 1$ : Proposition 2’s own bound,  
 105 retained verbatim rather than weakened to  $o(T)$ . Using condition (4) alone this already certifies  
 106  $\tau \leq b/2 + \inf_t \hat{q}_t$  and  $\tau < b/2 - \sup_t \hat{q}_t$ , which for a constant scorecaster is Corollary 2’s radius at  
 107  $\mu_t = 1$ .

108 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If  $\hat{q}_t \equiv \hat{q}$ , if  
 109  $\Lambda_t^\pm = A_t^\pm$ , and if neither depends on  $t$ , then  $\sigma^\pm = \tau^{*\pm}$  and **coverage is retained against every legal**  
 110 **adapted adversary if and only if  $\tau \leq \tau^{*+}$  and  $\tau < \tau^{*-}$ .** For a symmetric saturator that is one line:  
 111 coverage if and only if  $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$ , with the endpoint included only when  $\hat{q} < 0$ .  
 112 Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**  
 113 **before is the failure half.**  $\tau^{*+}$  is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs  
 114 agree with it read against the adversary the harness plays. It is not a retention condition, it omits  $\tau^{*-}$   
 115 entirely, and Table 2 (App. A) separates the two with legal objects only. The supremum can sit where  
 116 the dynamics never go: since  $|E_t| \leq (1 - \alpha)t$ , a saturator equal to the harness’s inside  $|x| \leq t^2$  and to  
 117  $\pm 10b$  outside it satisfies (4), reads  $\tau^{*+} = 19$  and fails at  $\tau = 1.001$ ; and a legal time-varying  $\hat{q}$  splits  
 118  $\sup_t$  from  $\inf_t$ ,  $\hat{q}_t = b/2$  on  $t$  a power of two and 0 elsewhere reading  $\tau^{*+} = 2$  against  $\sigma^+ = 1$ .

119 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric  
 120 saturator that meets (4) with equality and attains its extremes there ( $\Lambda_t^\pm = A_t^\pm = \pm b$ ; the harness’s  
 121  $r_t = b \text{ clip}(x/(ch(t)), \pm 1)$  is one, Section 2), and to a constant  $\hat{q}$ . Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

122 with the endpoint admissible only when  $\hat{q} < 0$ : exactly Corollary 2’s radius at  $\mu_t = 1$ , now as an  
 123 if-and-only-if. **The necessity direction above assumes none of it:**  $\tau > \tau^{*+}$ , or  $\tau \geq \tau^{*-}$ , forfeits  
 124 coverage against every admissible  $r_t$  and every legal predictable  $\{\hat{q}_t\}$ , extremes attained or not, and  
 125 that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather  
 126 than decorative: the unattained supremum and the varying- $\hat{q}$  split two paragraphs above are exactly

what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they always do.

**The mirror boundary is already measured, on this paper’s own grid.** The  $\tau$ -cliff run’s exact-threshold column, read against the mirror adversary  $s_t \equiv -b/2$  (Figure 2, App. A), locates  $\tau^{*-} = 1$  at the null scorecaster, closed on the failing side, exactly where (2) puts it. Both 1.000000 and 0.000000 are failures of the same order,  $|E_T| = (1 - \alpha)T$  and  $\alpha T$ , **so the failure criterion is  $|E_T|$ , not the miscoverage rate**; and once  $\hat{q}$  varies the rate stops being a tell at all.

**Both degenerate edges, and what they need.** An unbounded supremum is not on its own enough for  $\tau^{*+} = \infty$  to mean what it looks like: the same unreachable-set construction with  $r_t(x) = x$  past  $|x| > t^2$  has no finite supremum and fails at  $\tau = 1.5$ . What the tangent family has, and that construction does not, is a genuine reach:  $\Lambda_t(X) = \tan(\arctan(b)X/(c h(t)))$ , so the accumulator level at which the deployed threshold is forced to  $b/2$  is  $\kappa(\tau) c h(t)$  with  $\kappa(\tau) = \arctan(b/2 + \tau - \hat{q})/\arctan(b)$ , finite at every finite  $\tau$ . Coverage is retained at every width, as measured; but  $\kappa(\tau) > 1$  once  $\tau > b/2 + \hat{q}$ , so Proposition 2’s own *constant* is forfeited past  $\tau = b/2$  while coverage is not. At  $\tau = 1.5$ ,  $T = 10^6$  the excursion is 15.10 against Proposition 2’s 14.8155 and against  $\kappa(1.5) c h(T) + 1 = 15.8530$ . At the other edge,  $\sup_t \hat{q}_t = -b/2$  forces  $\hat{q} \equiv -b/2$ , since every  $\hat{q}_t$  is at least  $-b/2$  already. There  $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$  for every  $\tau > 0$  and every  $t$ , and  $\hat{q}_1 = 0 < b/2$ , so the failing set is absorbing from round 1 with no transient: the admissible window is the single point  $\tau = 0$ , and  $\tau = 0.5$  returns 1.000000, as does partial adjustment at  $w = 0.999$ .

**What that establishes is that the published condition is tight.** At  $\mu_t = 1$ ,  $\hat{q} \equiv 0$ ,  $\tau = 0.9$  covers at 0.100012 while  $\tau = 1.5$  returns 1.000000 with  $\max_t |E_t| = 900,000$ , 60,747 times Proposition 2’s bound at  $T = 10^6$  (Table 1, App. A); (2) makes that an if-and-only-if against the whole legal adversary class, not a pattern across five settings. **The claim is that necessity: a tightness result about someone else’s sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only, and now provably so: the tangent integrator above forfeits no coverage at any width Corollary 2 permits. At  $\hat{q} \equiv +b/2$  the looseness goes the other way and disappears: Corollary 2 permits radius 0 and (2) no width at all against the mirror adversary, the unsmoothed control included, though  $\tau = 1.5$  covers there against the specified one. That asymmetry makes “tight” a located claim.

**The boundary read as a bet.** The boundary has a betting reading (Appendix B): under a Bernoulli( $\alpha$ ) null on  $\text{err}_t$ , imposed for that reading alone and on nothing else here, an anytime-valid test convicts the  $\tau = 1.5$  arm past the edge at the 5% level and cannot reject inside it at any horizon. It adds a diagnostic and a reading, not a result, and no result above assumes that null or is certified by that test.

## 4 Limitations

**The retention direction is proved with no gap only where the two boundaries meet:** a constant scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run here; a genuinely time-varying  $\hat{q}$  leaves the band  $\sigma^+ < \tau \leq \tau^{*+}$  open in neither direction (Section 3). And the excursion is specific to the saturator’s *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator  $h(t) = t/\log t$ , the certified gap  $(c h(T) + 1)/T$  decays as  $O(1/\log T)$  rather than  $O(1/T)$ . What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.**  $\tau$  is in score units with  $b = 2$ , so the failing  $\tau = 1.5$  is  $0.75 b$ . Under i.i.d. scores it returns 0.249376 at  $T = 10^6$ , an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at  $b - \tau = 0.5$ , and uniform scores on  $[-b/2, b/2]$  put  $P(s > 0.5) = 0.25$  exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. The i.i.d. check above used one seed** (20260820,  $T = 10^6$ ); **the primary, adversarial harness uses none, and only its null-scorecaster edge is bracketed**; the others are one-sided. **What remains open is the band between the two boundaries.** For a genuinely time-varying  $\hat{q}$ , coverage in  $\sigma^+ < \tau \leq \tau^{*+}$  is neither proved nor falsified; Table 2’s power-of-two rows sit there, and closing that band is future work, not a sweep this paper has already run.

## References

- Lachlan L. H. Andrew, Siddharth Barman, Katrina Ligett, Minghong Lin, Adam Meyerson, Alan Roytman, and Adam Wierman. A tale of two metrics: Simultaneous bounds on competitiveness and regret. In *Proceedings of the 26th Conference on Learning Theory (COLT 2013)*, pages 741–763, 2013.
- Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/22000000101.
- Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. doi: 10.52202/075280-1000.
- Allen Emmanuel Binny and Anushri Dixit. Who moved my distribution? Conformal prediction for interactive multi-agent systems. arXiv:2511.11567v1, 2025.
- Jiadong Chen, Yang Luo, Xiuqi Huang, Fuxin Jiang, Yangguang Shi, Tieying Zhang, and Xiaofeng Gao. IPOC: An adaptive interval prediction model based on online chasing and conformal inference for large-scale systems. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 202–212, 2023. doi: 10.1145/3580305.3599396.
- George M. Constantinides. Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862, 1986. doi: 10.1086/261410.
- M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990. doi: 10.1287/moor.15.4.676.
- Ricarda Duerst, Jonas Schöley, Julia Hellstrand, and Mikko Myrskylä. Calibrating probabilistic forecast paths on past forecast errors: an application to the Finnish total fertility rate. Technical Report WP-2024-016, Max Planck Institute for Demographic Research, 7 2024.
- Théo Dupuy, Binbin Xu, Stéphane Perrey, Jacky Montmain, and Abdelhak Imoussaten. Relevance-aware thresholding in online conformal prediction for time series. In *Artificial Intelligence for Healthcare, and Hybrid Models for Coupling Deductive and Inductive Reasoning: First International Joint Conference, HC@AIxIA+HYDRA 2025, Bologna, Italy, October 25–26, 2025, Proceedings*, volume 2830 of *Communications in Computer and Information Science*, pages 196–217, 2026. doi: 10.1007/978-3-032-16708-8\_17.
- Uwe Ehret. Convergence index: a new performance measure for the temporal stability of operational rainfall forecasts. *Meteorologische Zeitschrift*, 19(5):441–451, 2010. doi: 10.1127/0941-2948/2010/0480.
- Sergio Galeani, Sophie Tarbouriech, Matthew Turner, and Luca Zaccarian. A tutorial on modern anti-windup design. *European Journal of Control*, 15(3-4):418–440, 2009. doi: 10.3166/ejc.15.418-440.
- Ben Gao, Jordan Patracone, Stéphane Chrétien, and Olivier Alata. Conformal online learning of deep Koopman linear embeddings. arXiv:2511.12760v2, 2025.
- Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.
- Evgenii Genov, Julian Ruddick, Christoph Bergmeir, Majid Vafaeipour, Thierry Coosemans, Salvador García, and Maarten Messagie. Balancing forecast accuracy and switching costs in online optimization of energy management systems. *Expert Systems with Applications*, 298:129305, 2026. doi: 10.1016/j.eswa.2025.129305.
- Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- Isaac Gibbs and Emmanuel J. Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 25(162):1–36, 2024.

- 225 Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration  
226 and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2):  
227 243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.
- 228 Rakshitha Godahewa, Christoph Bergmeir, Zeynep Erkin Baz, Chengjun Zhu, Zhangdi Song, Salvador  
229 García, and Dario Benavides. On forecast stability. *International Journal of Forecasting*, 41(4):  
230 1539–1558, 2025. doi: 10.1016/j.ijforecast.2025.01.006.
- 231 Dongjian Hu, Junxi Wu, Shu-Tao Xia, and Changliang Zou. Distribution-informed online conformal  
232 prediction. arXiv:2512.07770v2, 2026.
- 233 Ankit Lade, Sai Krishna J., and Indar Kumar. Bias-corrected adaptive conformal inference for  
234 multi-horizon time series forecasting. arXiv:2604.13253v1, 2026.
- 235 Ruipu Li and Alexander Rodríguez. Neural conformal control for time series forecasting. In  
236 *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 18439–18447,  
237 2025. doi: 10.1609/aaai.v39i17.34029.
- 238 Ruipu Li, Daniel Menacho, and Alexander Rodríguez. Optimization-based online conformal predic-  
239 tion for multi-step forecasting. arXiv:2508.13362v3, 2025.
- 240 Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, and Jiaye Teng. Questioning the coverage-length  
241 metric in conformal prediction: When shorter intervals are not better. arXiv:2601.21455v2, 2026.
- 242 F. Pappenberger, H. L. Cloke, A. Persson, and D. Demeritt. HESS Opinions “on forecast  
243 (in)consistency in a hydro-meteorological chain: curse or blessing?”. *Hydrology and Earth  
244 System Sciences*, 15(7):2391–2400, 2011. doi: 10.5194/hess-15-2391-2011.
- 245 P. Pinson and R. Girard. Evaluating the quality of scenarios of short-term wind power generation.  
246 *Applied Energy*, 96:12–20, 2012. doi: 10.1016/j.apenergy.2011.11.004.
- 247 Kandrika Pritularga and Nikolaos Kourentzes. Forecast congruence: A quantity to align forecasts  
248 and inventory decisions. SSRN working paper 4711817, 2024.
- 249 Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and  
250 safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.
- 251 Pouria Sarhadi. On the standard performance criteria for applied control design: PID, MPC or  
252 machine learning controller? arXiv:2503.14379v3, 2025.
- 253 Glenn Shafer and Vladimir Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wi-  
254 ley Series in Probability and Statistics. Wiley, 2019. ISBN 9780470903056. doi: 10.1002/  
255 9781118548035.
- 256 Huseyin Tunc, Onur A. Kilic, S. Armagan Tarim, and Burak Eksioğlu. A simple approach for  
257 assessing the cost of system nervousness. *International Journal of Production Economics*, 141(2):  
258 619–625, 2013. doi: 10.1016/j.ijpe.2012.09.022.
- 259 Jente Van Belle, Ruben Crevits, and Wouter Verbeke. Improving forecast stability using deep learning.  
260 *International Journal of Forecasting*, 39(3):1333–1350, 2023. doi: 10.1016/j.ijforecast.2022.06.  
261 007.
- 262 Jente Van Belle, Honglin Wen, Wouter Verbeke, and Pierre Pinson. Stabilizing distribution-free  
263 probabilistic forecasts. arXiv:2605.28531v1, 2026.
- 264 Rahul Vaze. Simultaneous coverage and efficiency guarantee in online conformal prediction.  
265 arXiv:2607.26577v1, 2026.
- 266 Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals  
267 of Statistics*, 49(3):1736–1754, 2021. doi: 10.1214/20-AOS2020.
- 268 Xiaoqian Wang and Rob J. Hyndman. Online conformal inference for multi-step time series forecast-  
269 ing. arXiv:2410.13115v2, 2024.

- 270 Rochelle P. Worsnop, Michael Scheuerer, Thomas M. Hamill, and Julie K. Lundquist. Generating  
271 wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-  
272 processing. *Wind Energy Science*, 3(1):371–393, 2018. doi: 10.5194/wes-3-371-2018.
- 273 Junxi Wu, Dongjian Hu, Yajie Bao, Shu-Tao Xia, and Changliang Zou. Error-quantified conformal  
274 inference for time series. arXiv:2502.00818v2, 2025.
- 275 Marco Zanotti. Analyzing the retraining frequency of global forecasting models: towards more stable  
276 forecasting systems. arXiv:2506.05776v3, 2026.
- 277 Mingyi Zheng, Hongyu Jiang, Yizhou Lu, and Jiaye Teng. Smoothing-based conformal prediction  
278 for balancing efficiency and interpretability. arXiv:2509.22529v1, 2025.
- 279 Ervin Zsótér, Roberto Buizza, and David Richardson. “Jumpiness” of the ECMWF and Met Office  
280 EPS control and ensemble-mean forecasts. *Monthly Weather Review*, 137(11):3823–3836, 2009.  
281 doi: 10.1175/2009MWR2960.1.

## 282 Code and data

283 The simulator, its frozen configuration, the results/ files behind every figure and table, the gen-  
284 erators that read them, and this project’s audit trail across sessions are at [https://\[REPOSITORY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])  
285 [LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION\]](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]).

## 286 A Supporting measurements

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to  $T = 10^6$ ; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at  $T = 2 \times 10^5$  and 14.8155 at  $T = 10^6$ ; “ratio” is  $\max_t |E_t|$  over it, and “Sat.” the fraction of rounds on which (4) delivers its full  $\pm b$ . Every number is measured against the adversary Section 2 specifies.

| Readout on the completed threshold | miscov. $2 \times 10^5$ | miscov. $10^6$  | $\max_t  E_t  10^6$ | ratio  | Sat. $10^6$ | travel $10^6$ |
|------------------------------------|-------------------------|-----------------|---------------------|--------|-------------|---------------|
| none (control)                     | 0.100035                | 0.100007        | 7.80                | 0.53   | 0.0000      | 28,311        |
| partial adj. $w = 0.5$             | 0.100030                | 0.100007        | 7.70                | 0.52   | 0.0000      | 18,113        |
| partial adj. $w = 0.9$             | 0.100030                | 0.100007        | 7.50                | 0.51   | 0.0000      | 4,081         |
| partial adj. $w = 0.99$            | 0.100040                | 0.100006        | 63.00               | 4.25   | 0.0007      | 1,023         |
| partial adj. $w = 0.999$           | 0.100035                | <b>0.100004</b> | 623.70              | 42.10  | 0.0097      | 202           |
| growing time constant $p = 0.5$    | 0.100030                | 0.100006        | 12.60               | 0.85   | 0.0000      | 330           |
| growing time constant $p = 0.9$    | 0.100030                | 0.100024        | 53.00               | 3.58   | 0.3181      | 16            |
| running mean                       | 0.099940                | 0.100010        | 49.50               | 3.34   | 0.3647      | 9             |
| dead band $\tau = 0.5$             | 0.100035                | 0.100005        | 10.60               | 0.72   | 0.0000      | 2,045         |
| dead band $\tau = 0.9$             | 0.100060                | 0.100012        | 13.20               | 0.89   | 0.0038      | 1,051         |
| dead band $\tau = 1.5$             | <b>1.000000</b>         | <b>1.000000</b> | 900,000             | 60,747 | 1.0000      | 0.5           |

Table 2: Legal  $(r_t, \hat{q})$  configurations on which the printed  $\tau^*$  over-certifies, each under two legal adversaries; “proved window” is where (2) certifies retention against every legal adversary, and between it and the failure boundaries  $\tau^{*\pm}$  neither direction is claimed, which is where the rows built on a supremum the dynamics cannot reach or never attain sit. Here  $\emptyset$  marks a configuration for which no width is certified,  $\tau = 0$  included, and  $T = 10^5$  except the two power-of-two rows, at  $T = 10^6$ . The last row carries no dead band at all.

| $(r_t, \hat{q})$ configuration        | $\tau$ | printed $\tau^*$ | proved window | $s_t = \hat{q}_t + \varepsilon$ |                | $s_t \equiv -b/2$ |                |
|---------------------------------------|--------|------------------|---------------|---------------------------------|----------------|-------------------|----------------|
|                                       |        |                  |               | miscov.                         | $\max_t  E_t $ | miscov.           | $\max_t  E_t $ |
| $\hat{q} \equiv +0.4$ , level $b$     | 0.599  | 1.4              | $\tau < 0.6$  | 0.100010                        | 7.20           | 0.099890          | 11.60          |
| $\hat{q} \equiv +0.4$ , level $b$     | 0.8    | 1.4              | $\tau < 0.6$  | 0.100050                        | 8.70           | <b>0.000000</b>   | 10,000         |
| $\sup_x r_t = 10b$ , off the reach    | 1.001  | 19               | $\tau < 1$    | <b>1.000000</b>                 | 90,000         | 0.000000          | 10,000         |
| $\sup_x r_t$ unbounded, off the reach | 1.5    | $\infty$         | $\tau < 1$    | <b>1.000000</b>                 | 90,000         | 0.000000          | 10,000         |
| $A^+ = 4b$ , $A^- = -b$               | 1.5    | 7                | $\tau < 1$    | 0.100020                        | 4.30           | <b>0.000000</b>   | 10,000         |
| level $4b$ , $\hat{q} \equiv 0$       | 6.999  | 7                | $\tau < 7$    | 0.100070                        | 12.20          | 0.100050          | 11.60          |
| level $4b$ , $\hat{q} \equiv 0$       | 7.0    | 7                | $\tau < 7$    | 0.100090                        | 12.20          | <b>0.000000</b>   | 10,000         |
| $\sup_x r_t = 2b$ , unattained        | 2.99   | 3                | $\tau < 1$    | 0.100320                        | 211.10         | 0.098010          | 210.50         |
| $\sup_x r_t = 2b$ , unattained        | 3.0    | 3                | $\tau < 1$    | <b>1.000000</b>                 | 90,000         | 0.000000          | 10,000         |
| $\sup_x r_t = 2b$ , attained          | 3.0    | 3                | $\tau < 3$    | 0.100000                        | 12.20          | <b>0.000000</b>   | 10,000         |
| $\hat{q}_t = b/2$ on powers of 2      | 1.0    | 2                | $\emptyset$   | 0.100009                        | 14.30          | <b>0.000000</b>   | 100,000        |
| $\hat{q}_t = b/2$ on powers of 2      | 1.9    | 2                | $\emptyset$   | <b>0.110960</b>                 | 84,745.70      | 0.000000          | 100,000        |
| $\hat{q} \equiv +b/2$ , no band       | 0      | 2                | $\emptyset$   | 0.100000                        | 0.90           | <b>0.000000</b>   | 10,000         |

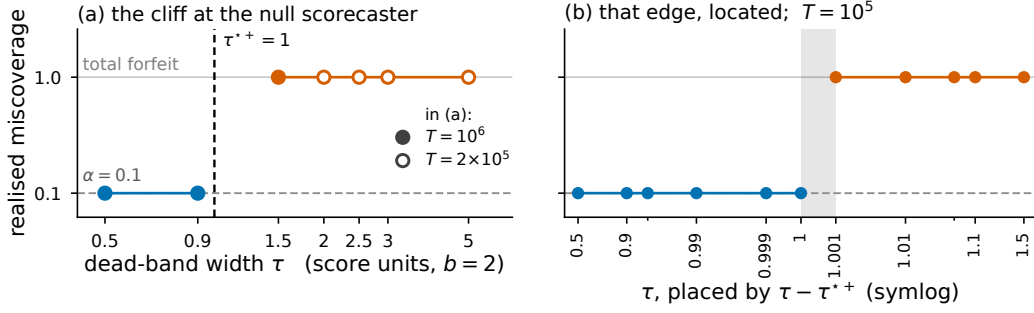


Figure 2: Realised miscoverage against dead-band width  $\tau$  at the null scorecaster,  $\hat{q} \equiv 0$  with a saturator of level  $b$ , under the *specified* adversary, where the upward boundary  $\tau^{*+}$  of (2) is 1: **(a)** the full sweep, filled markers to  $T = 10^6$  and the four widest bands, drawn open, to  $T = 2 \times 10^5$ , blue covering and orange forfeiting, with the dashed rule at  $\tau^{*+}$  and the grey band spanning the widest covering width to the narrowest failing one. **(b)** The same setting at  $T = 10^5$  over eleven measured widths, placed by  $\tau - \tau^{*+}$  on a symlog axis (linear zone  $|\tau - \tau^{*+}| \leq 10^{-3}$ ; unlabelled minor ticks at  $\tau = 0.95$  and  $1.05$ ) that magnifies the neighbourhood of  $\tau^{*+}$ , so visual spacing there is not proportional to true distance in  $\tau$ . Figure 3 gives the other four  $(r_t, \hat{q})$  settings.

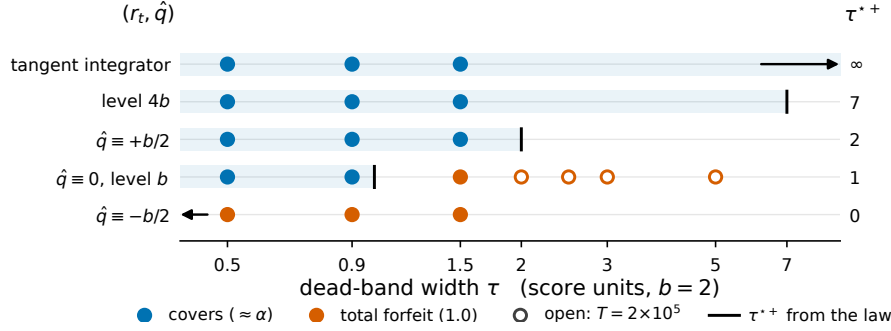


Figure 3: All 19 dead-band runs at the five admissible  $(r_t, \hat{q})$  settings, rows ordered by  $\tau^*$  on the shared horizontal  $\tau$  axis: blue markers cover at  $\approx \alpha$  and orange markers return 1.000000, filled to  $T = 10^6$  and open to  $T = 2 \times 10^5$ , all under the *specified* adversary. The rule on each row is that setting's upward boundary  $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ , printed at the right, with the tinted band  $\tau \leq \tau^{*+}$ ; two rules fall off the axis and are drawn as arrows,  $\tau^{*+} = 0$  at  $\hat{q} \equiv -b/2$  and  $\tau^{*+}$  unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary  $\tau^{*-}$  of (2) is not drawn here and differs from  $\tau^{*+}$  at the two non-null scorecasters.

## 287 B The betting reading

288 The boundary is stated in  $|E_T|$ , which is the net gain of a Skeptic who pays  $\alpha$  each round for  $\text{err}_t$ ;  
 289 that is what gives it a betting reading. The reading Section 3 points at is Ville's inequality applied to  
 290 the standard martingale  $M_t = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha))$ , whose logarithm depends on the path only  
 291 through  $t$  and  $E_t$  [Shafer and Vovk, 2019, Ramdas et al., 2023, Vovk and Wang, 2021], under the  
 292 imposed null that  $\text{err}_t \mid \mathcal{F}_{t-1}$  is Bernoulli( $\alpha$ ), with  $\mathcal{F}_{t-1}$  the history through round  $t - 1$ . Here  $\lambda$   
 293 is fixed in  $[-1/(1 - \alpha), 1/\alpha]$ , and its sign is chosen after the fact, not pre-registered. So read, an  
 294 anytime-valid test of the deployed intervals' calibration convicts the  $\tau = 1.5$  arm on its fifth interval  
 295 at the 5% level, and inside the edge the same test cannot reject at that level at any horizon.  $M_t$  is an  
 296 e-value for that imposed null alone; the accumulator itself is signed and is neither an e-value nor an  
 297 e-process, and no result in Section 3 assumes that null or is certified by that test.

## 298 C Where this sits

299 **One quantity, four names.** Four literatures name the movement of a sequentially updated forecast  
 300 in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC,  
 301 MQC/SMQC, congruence, nervousness); *jumpiness* [Zsótér et al., 2009], *convergence index* [Ehret,



2010], *(in)consistency* [Pappenberger et al., 2011]; and *switching cost*. The same arithmetic is the Multi-Quantile Change over a predictive distribution [Zanotti, 2026, v3, §3.4.1], IACER in applied control [Sarhadi, 2025], and a path length in dynamic-regret online learning. Online conformal prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al., 2023, Thm 1, Prop. 2], generalizing the running-error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold. Forecast stability treats the movement as a functional, read out as post-processing [Godahewa et al., 2025, Eq. 3], against a priced decision to switch [Genov et al., 2026, §4.2], [Van Belle et al., 2026]. Verification work uses a matched-level design against real producers from 2012 [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning gives worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2].

The object that settles a deployed interval’s validity (the admissible set around the deployed threshold) sits in the first of them. Those four vocabularies are where this paper’s claim and its concessions have to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that  $\sum_{t \leq T} q_t \leq O(\alpha h(T))$  for a sublinear  $h$ , supported with a plot rather than derived. Verification’s matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing  $\theta$  with  $T$ , which grows the ratio [Andrew et al., 2013, Thm. 7].